MAT2002 **Name (Print):** _____
Spring 2025
Midterm Exam
18.30-20.30, March 30, 2025
Time Limit: 120 Minutes

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1. (30 points) True or false?
   (a) $y'' = cos(t)y' + tsin(y)$ is a second order linear ODE.
   (b) $x^2y'' = y$ is an Euler equation, it has two solutions of the form $y_1 = x^{r_1}, y_2 = x^{r_2}$ for some constant r_1, r_2
   (c) The function e^{x^3} is an integrate factor for the ODE $y' - 3x^2y = x$
   (d) The general solution to $xy' - (y-1)^2 = 0$ is $y = \frac{c+ln|x|-1}{c+ln|x|}$
   (e) $y = 0$ is always a solution to a linear homogeneous ODE.
   (f) The set of solutions for a linear ODE forms a vector space.
   (g) The initial value problem $y' = x\tan(y),\ y(0) = 0$ has a unique solution.
   (h) A sequence of functions $\{\frac{1}{nt}\}_{n=1}^{\infty}$ converge uniformly to 0.
   (i) Suppose y_1 and y_2 are two fundamental solutions to the ODE $y'' = 2xy' + y$ then the Wronskian of y_1 y_2 is $W(y_1, y_2) = x^2$
   (j) If $p(x)$ and $q(x)$ are continuous functions on the interval (a, b), then the IVP $y' + p(x)y = q(x), y(x_0) = y_0$ always admits a unique solution for any $x_0 \in (a, b)$ and $y_0 \in \mathbb{R}$.

Solution: (a) F (b) T (c) F (d) F (e) T (f) F (g) T (h) F (i) F (j) T

2. (15 points) Solve the following equations:
   (a) $\frac{dy}{dx} = (y+1)^2\cos(2x)$.
   (b) $y'' - 2y' + 10y = 0$.
   (c) $\frac{dy}{dx} = \frac{x+y}{x-y}$.

Solution: '(a) (i) $y = -1$ is a solution;
(ii) if $y \neq -1$, then

$$\frac{dy}{(y+1)^2} = \cos(2x)dx$$
$$-\frac{1}{y+1} = \frac{1}{2}\sin(2x) + C$$
$$y = -1 - \frac{1}{\frac{1}{2}\sin(2x) + C}$$

So the general solution is $y = -1$ or $y = -1 - \frac{1}{\frac{1}{2}\sin(2x)+C}, C \in \mathbb{R}$.
(b) Consider the characteristic equation

$$r^2 - 2r + 10 = 0, \quad r_1 = 1 + 3i, r_2 = 1 - 3i.$$

The general is given by $y = C_1 e^t \cos(3t) + C_2 e^t \sin(3t)$.
(c) $\frac{dy}{dx} = \frac{1+\frac{y}{x}}{1-\frac{y}{x}} = \frac{1+v}{1-v}$
$x\frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v^2}{1-v}$
$\frac{1-v}{1+v^2}dv = \frac{1}{x}dx$
$\arctan v - \ln\sqrt{1+v^2} = \ln|x| + C$
Substitute back: $\sqrt{x^2+y^2} = ce^{\arctan\frac{y}{x}} (c > 0)$

3. (10 points) Consider

$$\left(4x^2y + 2xy + \frac{4}{3}y^3\right)dx + \left(x^2 + y^2\right)dy = 0.$$

(a) Find an integrating factor μ, such that the equation becomes an exact equation if we multiply it by μ.

(b) Solve the general solutions.

Solution: (a) Let $M = 4x^2y + 2xy + \frac{4}{3}y^3, N = x^2 + y^2$, then $M_y = 4x^2 + 2x + 4y^2, N_x = 2x$. Since $\frac{M_y - N_x}{N} = 4$ does not depend on y, let $\mu = \mu(x)$ be the integrating factor and satisfies

$$\frac{\mu'(x)}{\mu(x)} = 4,$$
$$\ln|\mu| = 4x$$

we choose $\mu = e^{4x}$, and multiply it to the original equation,

$$e^{4x}\left(4x^2y + 2xy + \frac{4}{3}y^3\right)dx + e^{4x}\left(x^2 + y^2\right)dy = 0$$

This must be an exact equation.
(b) Let

$$\begin{aligned}\Psi(x,y) &= \int e^{4x}\left(x^2 + y^2\right)dy + h(x) \\ &= e^{4x}\left(x^2y + \frac{1}{3}y^3\right) + h(x),\end{aligned}$$

then

$$\Psi_x = e^{4x}\left(4x^2y + 2xy + \frac{4}{3}y^3\right) + h'(x) = e^{4x}\left(4x^2y + 2xy + \frac{4}{3}y^3\right),$$

so

$$h'(x) = 0, h(x) = C_1.$$

The general solution is given by

$$e^{4x}\left(x^2y + \frac{1}{3}y^3\right) = C, C \in \mathbb{R}$$

4. (15 points)Solve the following :
   (a) $y'' + 2y' + y = (t-1)e^{-t}$
   (b) $y'' + 2y' + y = e^{-t}\ln t$

Solution: Solve the characteristic equation $r^2+2r+1 = 0, r_1 = r_2 = -1$. So $y_1 = e^{-t}, y_2 = te^{-t}$ form a fundamental set of solutions.
(a) Let a particular solution to the ODE be

$$\begin{aligned} & Y(t) = t^2e^{-t}(At+B), \\ \text{then} \quad & Y'(t) = e^{-t}\left(-At^3 + (3A-B)t^2 + 2Bt\right), \\ & Y''(t) = e^{-t}\left(At^3 + (-6A+B)t^2 + (6A-4B)t + 2B\right), \end{aligned}$$

Putting them into the equation, we get $e^{-t}(6At + 2B) = e^{-t}(t-1)$, so $A = \frac{1}{6}, B = -\frac{1}{2}$. Then the particular solution is

$$Y(t) = e^{-t}\left(\frac{1}{6}t^3 - \frac{1}{2}t^2\right)$$

Then the general solution to the ODE is

$$y(t) = C_1e^{-t} + C_2te^{-t} + e^{-t}\left(\frac{1}{6}t^3 - \frac{1}{2}t^2\right)$$

(b) Let $Y(t) = u_1y_1 + u_2y_2 = u_1e^{-t} + u_2te^{-t}$ be a particular solution. The Wronskian is $W(y_1, y_2) = e^{-2t}$. By variation of parameters,

$$\begin{aligned} u_1' &= -\frac{gy_2}{W} = -t\ln t \\ u_2' &= \frac{gy_1}{W} = \ln t \end{aligned}$$

Integration by parts, we solve

$$\begin{aligned} u_1 &= -\int t\ln t dt = t^2\left(-\frac{1}{2}\ln t + \frac{1}{4}\right) \\ u_2 &= \int \ln t dt = t(\ln t - 1) \end{aligned}$$

The particular solution is $Y(t) = u_1 e^{-t} + u_2 t e^{-t} = t^2 e^{-t}\left(\frac{1}{2}\ln t - \frac{3}{4}\right)$. Then the general solution to the ODE is

$$y(t) = C_1 e^{-t} + C_2 t e^{-t} + t^2 e^{-t}\left(\frac{1}{2}\ln t - \frac{3}{4}\right).$$

5. (15 points)Suppose the linear homogeneous equation

$$(t-1)y'' - ty' + y = 0, \quad t > 1$$

has one solution $y_1 = t^m$, where m is a positive integer. Find the value of m, and find another solution $y_2(t)$ such that $\{y_1, y_2\}$ form a fundamental set of solutions.

Solution: We substitute $y_1 = t^m$ into the equation, then we have

$$m(m-1)(t-1)t^{m-2} - mt^m + t^m = 0$$

so

$$(m-1)\left(-t^m + mt^{m-1} - mt^{m-2}\right) = 0, \quad t > 1.$$

It suggests that $m = 1$ and $y_1 = t$.
To find y_2, we have two methods.

Method 1: Abel's theorem.

Rewrite the equation into

$$y'' - \frac{t}{t-1}y' + \frac{1}{t-1}y = 0, t > 1.$$

By Abel's Theorem (choose the constant $c = 1$),

$$\begin{aligned} W(y_1, y_2)(t) &= \exp\left(-\int p(t)dt\right) \\ &= \exp\left(\int \frac{t}{t-1}dt\right) \\ &= \exp\left(\int 1 + \frac{1}{t-1}dt\right) \\ &= e^t(t-1). \end{aligned}$$

So

$$ty_2' - y_2 = e^t(t-1)$$
$$y_2' - \frac{1}{t}y_2 = \frac{e^t(t-1)}{t}.$$

Let $\mu(t) = e^{-\int \frac{1}{t}dt} = \frac{1}{t}$, then

$$\begin{aligned} y_2 &= t\int \frac{e^t(t-1)}{t^2}dt \\ &= t\int e^t(t-1)d\left(-\frac{1}{t}\right) \\ &= t\left(-\frac{e^t(t-1)}{t} + \int e^t dt\right) \\ &= e^t \end{aligned}$$

It is easy to check y_2 is another solution, and y_1, y_2 form the fundamental set of solutions because the Wronskian is non-zero.

Method 2: Reduction of order by variation of parameter

Let $y_2(t) = v(t)y_1(t) = tv(t)$ be another solution. Then

$$y_2' = tv'(t) + v(t),$$
$$y_2'' = tv''(t) + 2v'(t).$$

Substitute y_2 into the equation, one has

$$
\begin{aligned}
(t-1)y_2'' - ty_2' + y_2 &= 0,\\
t(t-1)v''(t) + \left(-t^2+2t-2\right)v'(t) &= 0,\\
\frac{v''}{v'} &= \frac{t^2-2t+2}{t(t-1)} = 1 + \frac{1}{t-1} - \frac{2}{t},\\
v' &= C\frac{e^t(t-1)}{t^2},\\
v &= C\int \frac{e^t(t-1)}{t^2}dt,\\
v &= C\int e^t(t-1)d\left(-\frac{1}{t}\right),\\
v &= C\left(-\frac{e^t(t-1)}{t} + \int e^t dt\right),\\
v &= C\frac{e^t}{t}.
\end{aligned}
$$

Choosing $C = 1, y_2 = tv(t) = e^t$. It is easy to check y_2 is another solution, and y_1, y_2 form the fundamental set of solutions because the Wronskian is non-zero.

6. (15 points) Consider the curve C on the xy-plane. It passes through the point $(1, 1)$. For any point $P = (x, y)$ on C, suppose the tangent line to C at P intersects the y-axis at point A and $|OA| = |AP|$. Find the equation of the curve C.

Solution: Let point $P(x, y)$ be an arbitrary point on the curve. Let the point A be $(0, a)$. The slope of AP is equal to the slope of the tangent to the curve at P. Then

$$
\begin{aligned}
\frac{a-y}{0-x} &= \frac{dy}{dx}\\
a &= y - x\frac{dy}{dx}.
\end{aligned}
$$

So $A = \left(0, y - x\frac{dy}{dx}\right)$.

Since $|OA| = |AP|$,

$$
\begin{aligned}
\left|y - x\frac{dy}{dx}\right| &= \sqrt{x^2 + \left(x\frac{dy}{dx}\right)^2}\\
y^2 - 2xy\frac{dy}{dx} &= x^2
\end{aligned}
$$

Therefore we have IVP

$$
\begin{cases}
\frac{dy}{dx} = \frac{y^2-x^2}{2xy}\\
y(1) = 1
\end{cases}
$$

We then have two ways to solve the IVP.
Method 1.
Let $v(x) = \frac{y}{x}$. Then $y = vx$ and so $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substitute to the equation and simplify, we have

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{v^2 - 1}{2v}, \\ x\frac{dv}{dx} &= -\frac{v^2 + 1}{2v}, \\ \frac{2v}{v^2 + 1}dv &= -\frac{1}{x}dx, \\ \ln\left(v^2 + 1\right) &= -\ln|x| + C_1, \\ \ln\left(\left(\frac{y}{x}\right)^2 + 1\right) &= -\ln|x| + C_1, \\ \left(\frac{y}{x}\right)^2 + 1 &= \frac{C}{|x|}. \end{aligned}$$

Since $y(1) = 1$, we have $C = 2$. Hence we have $\left(\frac{y}{x}\right)^2 + 1 = \frac{2}{|x|}$. Note that x must be positive. So the curve is given by $\left(\frac{y}{x}\right)^2 + 1 = \frac{2}{x}$.
Method 2. Rewrite the IVP as

$$\begin{cases} \left(x^2 - y^2\right)dx + 2xydy = 0 \\ y(1) = 1 \end{cases}$$

Let $M = x^2 - y^2, N = 2xy$, then $M_y = -2y, N_x = 2y$. Since $\frac{M_y - N_x}{N} = -\frac{2}{x}$ does not depend on y, let $\mu = \mu(x)$ be the integrating factor and satisfies

$$\begin{aligned} \frac{\mu'(x)}{\mu(x)} &= -\frac{2}{x} \\ \ln|\mu| &= -2\ln|x|, \end{aligned}$$

we choose $\mu = x^{-2}$, and multiply it to the original equation,

$$\left(1 - \frac{y^2}{x^2}\right)dx + \frac{2y}{x}dy = 0$$

This must be an exact equation. Let

$$\begin{aligned} \Psi(x, y) &= \int \frac{2y}{x}dy + h(x) \\ &= \frac{y^2}{x} + h(x), \end{aligned}$$

then

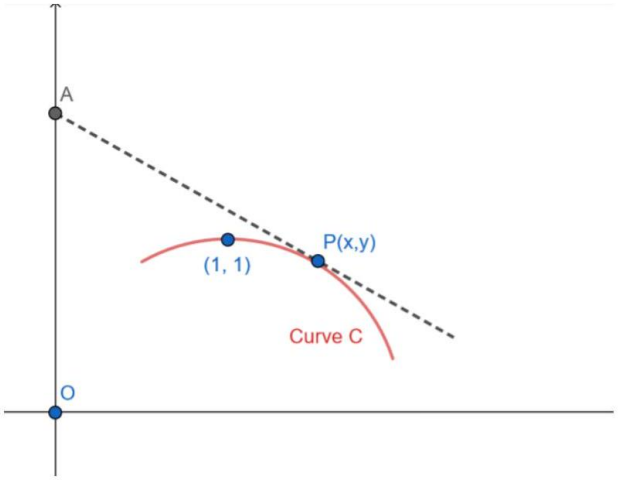


Figure 1: Question 6

$$\Psi_x = -\frac{y^2}{x^2} + h'(x) = 1 - \frac{y^2}{x^2},$$

so

$$h'(x) = 1, h(x) = x + C_1.$$

The general solution is given by

$$\frac{y^2}{x} + x = C, C \in \mathbb{R}.$$

Since $y(1) = 1$, we have $C = 2$. So the curve is given by $\frac{y^2}{x} + x = 2$.